End-to-End AI Video Studio Architecture

# Shine — Next-Gen AI Micro-Drama Studio

Transforming raw text prompts into viral 9:16 vertical drama series (20–50 episodes) with multi-agent AI directors, consistent character personas, in-browser WebGL editing, and Google Cloud serverless infrastructure.

MARKET SEGMENT
**Vertical Micro-Drama**
$10B+ Global Market Boom

CREATIVE BRAIN
**Multi-Agent AI Director**
Gemini 3.5 Flash & Veo 3.1

**Editor Core**
**Pixi.js WebGL NLE**
In-Browser Zero-Cost Render

INFRASTRUCTURE
**Cloud Run Serverless**
Scale-to-Zero ($0 Idle Cost)

MARKET CONTEXT

# The Explosive Rise of Vertical Micro-Dramas

## $10B+

**GLOBAL MARKET BY 2027**

Pioneered by Chinese Wēi Duǎnjù and rapidly expanding across the US, Southeast Asia, Latin America, and Europe via TikTok, ReelShort, DramaBox, and Shorts.

## 1–3 Min

**EPISODIC FORMULA**

Serialized 20–50 episode dramas with fast conflict escalation, intense emotional payoffs, and cliffhangers every 60–90 seconds to maximize viewer retention.

## 3.5x

**HIGHER MONETIZATION VELOCITY**

Micro-dramas convert viewers into paying subscribers 3.5x faster than traditional streaming platforms through pay-per-episode paywalls.

📊 **Opportunity:** The creator economy needs automated software to produce hundreds of high-quality vertical drama episodes per month with minimal budget.

MARKET PAIN POINTS

# Core Production Bottlenecks in Vertical Drama

### 1. High Physical Production Costs

Traditional filming requires casting actors, renting sets, filming crews, and extensive editing suites, costing $30,000–$100,000+ per series.

### 2. Character Visual Drift in Generative AI

Existing text-to-video tools fail to preserve consistent facial geometry, attire, and lighting across consecutive scenes, ruining narrative immersion.

### 3. Prohibitive Localization & Dubbing Costs

Translating video series into foreign languages traditionally requires re-rendering video clips from scratch, incurring massive GPU compute bills.

### 4. Disconnected Toolchains

Creators waste hours juggling separate tools for scriptwriting (ChatGPT), image gen (Midjourney), video gen (Runway), and NLE editors (Premiere).

TARGET MARKET

# Product-Market Fit & Ideal Customer Profiles

### Independent Writers & Creators

Web novel authors and solo creators who want to transform written stories into monetizable vertical video series without camera crews.

`Value: 100x faster production`

### Digital Media Studios & Agencies

Agencies producing high-volume serialized content for TikTok, YouTube Shorts, and Reels to capture ad revenue and subscriber paywalls.

`Value: 80% cost reduction`

### Global Localization Teams

Publishers localizing domestic drama hits into international markets (EN, VI, ZH, JA, KO, ES) with instant neural voice swapping.

`Value: Zero-video-render dubbing`

🎯 **The Shine Value Proposition:** Turn a single creative individual or small team into a full-fledged serialized drama production studio.

SYSTEM BLUEPRINT

# High-Level System Architecture & 5-Tier Pipeline

### 1. Client WebGL Studio
ACTIVE

Vue 3 SPA + Pixi.js v8 WebGL Canvas NLE. Zero-render real-time scrub & in-browser WebCodecs MP4 export.

### 2. Cloud API Gateway
ACTIVE

Node 22 Express :3001 + JWT Auth & 31 REST Controllers orchestrating multi-agent pipelines.

### 3. AI Director Brain
ACTIVE

Gemini 3.5 Flash & Veo 3.1 + Parallel AI Search MCP & Grafana MCP for viral trends & telemetry.

### 4. Cloud Serverless Workers
ACTIVE

Cloud Run Playwright Compositor + Meta Demucs v4 AI vocal isolation. Scale-to-Zero ($0 idle).

### 5. Data & Storage
ACTIVE

Firestore Native (shine-db), GCS Signed URLs & Google SynthID steganographic watermarking.


PROTOCOL: REST (HTTPS) • SSE Progress • MCP Protocols

**100% Google Cloud Native**

LAYER 1: PRESENTATION (ACTIVE CORE)

# OpenVideo Pixi.js WebGL & WebCodecs Engine

### 1. Real-Time WebGL Compositor (Pixi.js v8)

The browser viewport renders 9:16 vertical frames at $\ge 30\text{ FPS}$ using client GPU hardware.

- **Zero-Render Preview:** Scrub playhead across video, audio, and subtitle buffers with 0ms cloud latency.
- **GLSL Fragment Shaders:** Real-time chroma keying, color grading, and dynamic transition blends.
- **Microsecond Timeline Model:** Sub-millisecond timing precision ($1\text{s} = 1,000,000\mu\text{s}$).

### 2. In-Browser Export (WebCodecs API)

Renders MP4 video directly on the user's computer via hardware-accelerated WebCodecs (`mediabunny`).

- **Zero Server Cost ($0):** 100% computed on client device, saving thousands in GPU hosting bills.
- **Lightning Speed:** Encodes a 90-second 1080p episode in about 12 seconds.
- **In-Memory ISOBMFF Muxing:** Assembles the final MP4 container in memory for immediate download.

PLANNED ROADMAP SPECIFICATION (PHASE 2)

# Real-Time Team Collaboration Architecture

### Delta Patch Protocol (`PatchSyncService`)

Designed to broadcast atomic RFC 6902 delta patches ($<1\text{KB}$) over Socket.io instead of heavy 500KB blobs:

```
{ "op": "update", "path": "/clips/clip_01/timing/display/to", "value": 4500000, "author": "usr_sarah" }
```

Planned for co-editing sessions with sub-50ms peer update targets.

### Planned Concurrency Controls

- **Clip-Level Mutex Locking:** Optimistic locking to prevent race conditions when multiple users edit the same clip.
- **Live Teammate Cursors:** Playhead markers broadcasting peer listening positions at 30Hz.
- **Immutable Version Checkpoints:** Continuous timeline revision snapshots saved to Firestore.

Status: Architecture & Backend Service Designed ➔ Full Client Integration in Phase 2.

LAYER 3: AI BRAIN (ACTIVE CORE)

# Hierarchical Multi-Agent AI Director

### 1. Discovery & Skeleton

**TrendRadar:** Scans viral drama trends via Parallel MCP.
**StorySkeleton:** Designs 20–50 episode narrative master plans with cliffhangers.

`Model: gemini-3.5-flash-lite`

### 2. Screenplay Breakdown

**ScriptAgent:** Breaks each episode into 15–45 short scene blocks (4–8s) with visual prompts, actions, and dialogues in JSON.

`Model: gemini-3.5-flash-lite`

### 3. QA & In-Editor Copilot

**SupervisionAgent:** Audits pacing and safety.
**ChatbotAgent:** Modifies timeline state via natural language commands.

`Model: gemini-3.5-flash-lite / 2.5-pro`

🧠 **4-Tier Memory Mesh:** Active Session State ➔ Vertex Vector Search RAG (`text-embedding-004`) ➔ Series Knowledge Graph Lineage ➔ Token Compressor (compresses 500KB JSON into ~4KB Markdown prompts).

LAYER 3: LOCALIZATION (ACTIVE CORE)

# Decoupled Audio-Visual Dubbing Engine

### Track 1: Pure Visual Motion (VIDEO 1)

Generated as silent video clips via Google Veo 3.1 (`veo-3.1-generate-001`). Immutable during multi-language dubbing operations.

```
[Scene 01: 6.00s] —— [Scene 02: 5.50s] (Fixed Visuals)
```

### Track 2: Neural TTS Dialogue (AUDIO 1)

Synthesized per language track via `gemini-3.1-flash-tts-preview` (30 distinct voices) with word-level phonetic timestamps.

```
EN: "Where are you?" (2.1s) ➔ VI: "Anh đang ở đâu?" (2.6s, Δt)
```

💡 **Unfair Cost Advantage:** To distribute in 5 languages, Shine only swaps the voice track and recalculates timeline timings (\$\Delta t_{\mu s}\$). **You never re-render expensive video AI clips!**

LAYER 4: CLOUD WORKERS (ACTIVE CORE)

# Serverless Cloud Run Video Render Worker

**1. Asynchronous Jobs**
Episode render tasks are dispatched asynchronously (`/render-job`), tracking progress from 0% to 100% via API polling or SSE stream.

**2. Headless Playwright**
`shine-render-worker` container launches headless Chromium WebCodecs compositors to render master-quality 1080p/4K video frames.

**3. Scale-to-Zero ($0 Idle)**
Cloud Run containers scale automatically to 0 instances when idle, eliminating static server compute expenses.

🔍 **Parity Audit:** Automated SSIM diff tool ensures the final cloud video matches the editor preview with **SSIM > 0.999 parity**.

LAYER 4: AUDIO AI (ACTIVE CORE)

# Meta Demucs v4 AI Stem Separation & 3D Audio

### FastAPI Demucs v4 AI Microservice

A dedicated FastAPI Python microservice on Cloud Run (`services/demucs-worker`) separates mixed audio into isolated stems:

- **Vocal Isolation:** Separates character dialogue from background ambient noise.
- **Music Extraction:** Strips vocal tracks to create clean instrumental soundtracks.
- **Parametric Auto-Ducking:** Attenuates BGM by 80% with 250ms attack during dialogue intervals.

### 3D Binaural Spatial Audio Engine

Calculates real acoustic physics based on where characters stand on the phone screen:

- **Sabine RT60 Decay:** Simulates realistic room reverb (hallway, bedroom, street).
- **Woodworth Head Model:** Computes Interaural Time Differences (ITD) between ears.

```
Cinema-quality surround sound directly through standard headphones.
```

LAYER 5: STORAGE & TRUST (ACTIVE CORE)

# Pluggable Persistence, SynthID & C2PA Provenance

### Pluggable Database Architecture (`IDatabaseProvider`)

Enforces strict repository interfaces, enabling database switching via environment variables:

```
Firestore Native (`shine-db`) — Primary Cloud NoSQL
```

```
MongoDB Atlas — Document Database
```

```
SQLite (`shine.db`) — Local Development
```

### Digital Watermarking & Provenance

- **Google SynthID:** Embeds imperceptible digital watermarks into synthesized audio waveforms and video frames.
- **C2PA Content Credentials:** Cryptographically signs provenance manifests declaring AI generative origin.
- **V4 Signed URLs:** Generates temporary 30-minute GCS links for secure media streaming.

INFRASTRUCTURE SYNERGY

# Google Cloud Unified Service Synergy

### Compute & Scaling

**Cloud Run:** Serverless execution with `--min-instances 0` scaling to 0 when idle.

**Cloud Scheduler:** Periodic cron (`*/5 * * * *`) token sync heartbeat.

`$0 Idle Compute Cost`

### Data & Messaging

**Firestore Native:** Document persistence for scripts, timelines & version snapshots.

**Cloud Storage (GCS):** V4 signed URLs for high-throughput media streaming.

`gs://shine-studio-media`

### Security & Observability

**Cloud IAM & Secret Manager:** Least-privilege roles (`aiplatform.user`, `datastore.user`).

**Cloud Logging:** Real-time structured telemetry and P99 latency tracking.

`Enterprise IAM & Audit Trails`

🔗 **Synergy Advantage:** Native identity & networking across all GCP services eliminates ingress/egress transit bottlenecks and simplifies DevOps automation.

GENERATIVE AI STACK

# Multimodal Generative AI Model Matrix

| Domain / Task | Active Production Model | Supported Models & Fallbacks | Technical Role in Shine |
|---|---|---|---|
| **Script & Planning** | **`gemini-3.5-flash-lite`** | `gemini-2.5-pro`, `gemini-3.1-pro` | Story master plan, 15-45 scene JSON breakdown, cliffhangers. |
| **Agent Orchestration** | **`gemini-3.5-flash-lite`** | Multi-Agent ADK Router | TrendRadar, ScriptAgent, SupervisionAgent pipeline coordinator. |
| **Video Generation** | **`veo-3.1-generate-001`** | `veo-3.0`, `veo-2.0`, `veo-2.1` | 9:16 vertical silent video motion clips with camera trajectory steering. |
| **Facial Consistency** | **`gemini-3.1-flash-lite-image`** | `imagen-3.0-generate-002`, `imagen-3.5` | 8-angle facial consistency keyframes and character visual DNA. |
| **Neural TTS Speech** | **`gemini-3.1-flash-tts-preview`** | 30 Gemini Voices (`Zephyr`, `Puck`...) | Multi-speaker dialogue with word-level phonetic timestamps in 6 languages. |
| **Live Audio Dialogue** | **`gemini-live-2.5-flash-native-audio`** | Bidirectional WebSocket Stream | Low-latency live interactive character voice conversations & real-time dubbing. |
| **Soundtrack & SFX** | **`lyria-3-clip-preview`** | `Lyria-v1` | Dynamic mood-adaptive background music & suspense crescendo risers. |
| **Vector Search (RAG)** | **`text-embedding-004`** | Vertex Vector Search | Sub-50ms semantic search across scene bibles and character lineage. |
| **Stem Separation** | **`Meta Demucs v4 (htdemucs)`** | FastAPI PyTorch Container | Vocal isolation, BGM extraction, and parametric auto-ducking. |
| **Watermarking** | **`Google SynthID`** | C2PA Manifest Signing | Steganographic imperceptible watermarking on generated waveforms. |

PRIMARY API: Official `@google/genai` Node.js SDK

FALLBACK: Google Flow Session Account Pool (flowST)

PROTOCOL INTEGRATION

# Model Context Protocol (MCP) Integration Fabric

### 1. Parallel AI Search MCP (`search.parallel.ai/mcp`)

ACTIVE

Connects Shine's multi-agent director to real-time web search for discovering viral drama tropes and performing safety audits:

- **Trend Radar Agent:** Scans real-time trending micro-drama tropes across 6 regions.
- **Supervision Agent:** Automated pre-flight copyright check on titles, character names, and novel synopses.
- **SfxService:** Discovers open sound effects across Freesound, Pixabay, and web sources.

```
Client: `server/src/integrations/mcp/ParallelMCPClient.ts`
```

### 2. Grafana MCP & Loki Observability

ACTIVE

Unifies live system monitoring, container health, and error tracing through the official Grafana MCP endpoint:

- **Telemetry Streaming:** Auto-flushes structured logs and memory RSS (MB) to Grafana Cloud Loki every 15s.
- **P99 Latency & Probes:** Exposes `GET /api/admin/observability` tracking active WebSocket connections.
- **Automated Alerts:** Triggers instant email alerts upon critical AI pipeline failures.

```
Service: `server/src/services/observability/GrafanaObservabilityService.ts`
```

🔌 **Standardized Tool Calling:** Open MCP protocol ensures modular tool execution across Google Agent ADK without vendor lock-in.

COMPETITIVE EDGE

# Shine’s 4 Unique Selling Points (USPs)

### 1. Decoupled Multi-Language Dubbing

Silent visual clips + neural speech allow swapping dialogue into 6+ languages and auto-aligning timeline bounds ($\Delta t_{\mu s}$) without re-rendering expensive video AI clips.

### 2. In-Browser $0 GPU WebCodecs Render

Creators export 1080p MP4 episodes directly in browser memory in ~12 seconds, delivering fast turnarounds with zero server GPU overhead.

### 3. Persona Studio & 8 Facial Anchors

Locks facial geometry, wardrobe, and character embeddings across 50 episodes, eliminating visual drift and inconsistency.

### 4. Scale-to-Zero Serverless Economy

100% serverless infrastructure on Google Cloud Run automatically scales containers to 0 instances when idle, ensuring $0 static maintenance costs.

CONSISTENCY ENGINE

# Persona Studio: 8 Facial Consistency Anchors

### 8-Angle Facial Consistency Locking

When a character is registered, Persona Studio generates 8 multi-angle reference keyframes to lock their biometric DNA:

- Frontal Portrait
- 45° Left Profile
- 45° Right Profile
- Full Side Profile
- High Angle Shot
- Low Angle Cinematic
- Emotional Smirk
- Dramatic Anger

### Prompt DNA & Reference Injection

During video and storyboard generation, the character's visual traits and reference image URLs are automatically injected into Gemini and Veo prompts.

- Preserves signature hairstyles, scars, and attire continuity.
- Links voice IDs (Gemini Audio) to specific character personas.

Guarantees cast continuity across all 20-50 episodes.

AUDIENCE RETENTION

# Kinetic Subtitles & Dynamic Cliffhanger Engine

### Word-Level Animated Kinetic Captions

Captions drive over 60% of short-form video watch time. Shine automatically generates viral animated subtitles:

- **Karaoke Pop-In:** Active words highlight and bounce on beat with speech.
- **Typography Presets:** Dramatic Punch, Neon Cyberpunk, Clean Minimalist.
- **One-Click Translation:** Translates subtitles into foreign languages while preserving millisecond timings.

### Dynamic Cliffhanger Engine

Automatically identifies the episode climax and injects an intense 3-second hook sequence:

- **GLSL Shaders:** Triggers glitch, flash, or keyframe zoom-in effects.
- **Audio Stinger:** Injects dramatic crescendo risers on `AUDIO 3`.
- **Call-to-Action Overlay:** Displays high-impact prompt: *"Episode 2 Unlocked in 3s"*.

```
Maximizes episode completion rates and subscription paywall conversion.
```

DEVELOPMENT EVOLUTION

# Agile Engineering Roadmap (Sprints 1–7)

| | |
|---|---|
| **Sprint 1:** Foundation, Vertex AI SDK, Pluggable DBs (Firestore/SQLite), and Auth Suite. | v0.1 Alpha |
| **Sprint 2:** Multi-Agent Script Pipeline, Trend Radar (Parallel MCP), Persona Studio. | v0.2 Beta |
| **Sprint 3:** OpenVideo Pixi.js WebGL Editor, Undo/Redo Commands, Dual-Render Parity. | v0.5 NLE Beta |
| **Sprint 4:** Neural Voiceover, Kinetic Subtitles, Cliffhanger Engine, Meta Demucs v4. | v0.8 RC1 |
| **Sprint 5:** WebSocket Delta Sync Architecture, Quality Linter & i18n Key Parity. | v0.9 RC2 |
| **Sprint 6 & 7:** Multi-Platform Publishing, Grafana MCP Observability & Playwright E2E Suite. | v1.1 Commercial |

FUTURE PLANNING (PHASE 2 & 3 ROADMAP)

# Planned Roadmap Innovations

### 1. Real-Time Multi-User Co-Editing

Full client integration of the Socket.io RFC 6902 JSON Pointer delta patch engine for live collaborative timeline editing.

`Phase 2: Live Team Collaboration`

### 2. Distributed Batch Render Queue

Automated multi-episode batch queueing across a pool of parallel Cloud Run Playwright worker nodes via Cloud Pub/Sub.

`Phase 2: Automated 50-Episode Batch Queue`

### 3. Platform Analytics & Synthetic Audience

Viewer retention heatmaps, drop-off prediction curves, and cross-platform publishing analytics (TikTok / Shorts / Reels).

`Phase 3: Platform Analytics & Audience Simulator`

### 4. Live Interactive Drama & Marketplace

Branching narrative trees (`@antv/g6`) for live voting + AI character asset trading marketplace.

`Phase 3: Branching Drama & Character Hub`

COST ARCHITECTURE

# Cloud Run Sizing & $0 Idle Cost Model

| Service | Specs | Autoscaling | Concurrency | Idle Cost |
| --- | --- | --- | --- | --- |
| **`shine-app`** | 2 vCPU / 4Gi | 0 to 3 instances | 80 req/instance | **$0.00** |
| **`shine-render-worker`** | 4 vCPU / 8Gi | 0 to 3 instances | 1 job (Isolated) | **$0.00** |
| **`demucs-worker`** | 2 vCPU / 4Gi | 0 to 3 instances | 2 req/instance | **$0.00** |

💡 **Scale-to-Zero Guarantee:** When no users are actively editing or rendering, all 3 Cloud Run services scale to 0 instances. Zero monthly server bills when idle!

DEVOPS & AUTOMATION

# 1-Click Full Ecosystem Cloud Run Deployment

### Automated Cloud Provisioning Lifecycle

1. Auto-enables 10 GCP APIs (`run`, `pubsub`, `firestore`, `aiplatform`, etc.).
2. Grants least-privilege IAM roles to Default Compute Service Account.
3. Auto-provisions Firestore Native database `shine-db` and GCS bucket.
4. Creates Pub/Sub topics (`shine-render-jobs`, `shine-render-status`).
5. Builds and deploys all 3 containers with dynamic YAML configuration.

### Single Execution Command

```
# Windows PowerShell .\scripts\deploy-cloudrun.ps1 # Linux / macOS /
Cloud Shell ./scripts/deploy-cloudrun.sh
```

```
Health Check: `curl https:///api/health`
```

Strategic Summary

# Redefining Global Storytelling for the Short-Form Video Era

Shine unites cutting-edge Google generative AI models (`gemini-3.5-flash-lite`, `veo-3.1`, `gemini-live-2.5`, `lyria-3`), Model Context Protocol integrations (Parallel MCP & Grafana MCP), and serverless scale-to-zero infrastructure to turn any creator into a global micro-drama studio.

| 100x Faster | 80% Cheaper | $0 Idle Cost |
|---|---|---|
| Script to Video | Zero-Render Dubbing | Serverless Run |